

**COMMUNITY-ACQUIRED PNEUMONIA
& ACUTE RHINOSINUSITIS:**
A CLINICAL COMPENDIUM

EPIDEMIOLOGY & DISEASE BURDEN

Community-Acquired Pneumonia (CAP) and Acute Rhinosinusitis (ARS)

Community-Acquired Pneumonia — Global & India Burden

CAP is a leading cause of hospitalization and mortality, ranging from mild outpatient illness to severe ICU-requiring disease.¹

1.5–14 cases per 1,000 persons annually (global incidence) ¹	23% India's share of global pneumonia burden ²	36% WHO SE-Asia regional burden (<5 yrs) ²	14–30% Case fatality rate (all CAP) ²	≤47% Mortality in severe CAP ²
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- Ranked as the 8th leading cause of death overall and the leading infectious cause of death worldwide.¹
- 783,000 deaths reported from lower respiratory infections in South-East Asia (WHO, 2016).²

Acute Rhinosinusitis (ARS)

Acute rhinosinusitis is defined as inflammation of the paranasal sinuses and nasal cavity lasting ≤4 weeks, characterized by purulent nasal discharge with nasal obstruction and/or facial pain/pressure.³

BURDEN³

- ❑ Occurs in **15–40 episodes per 1,000 patients/year**
- ❑ **~0.5% of URTIs** are complicated by sinusitis
- ❑ Primarily **viral and self-limiting**
- ❑ Second most common infectious condition encountered in general practice

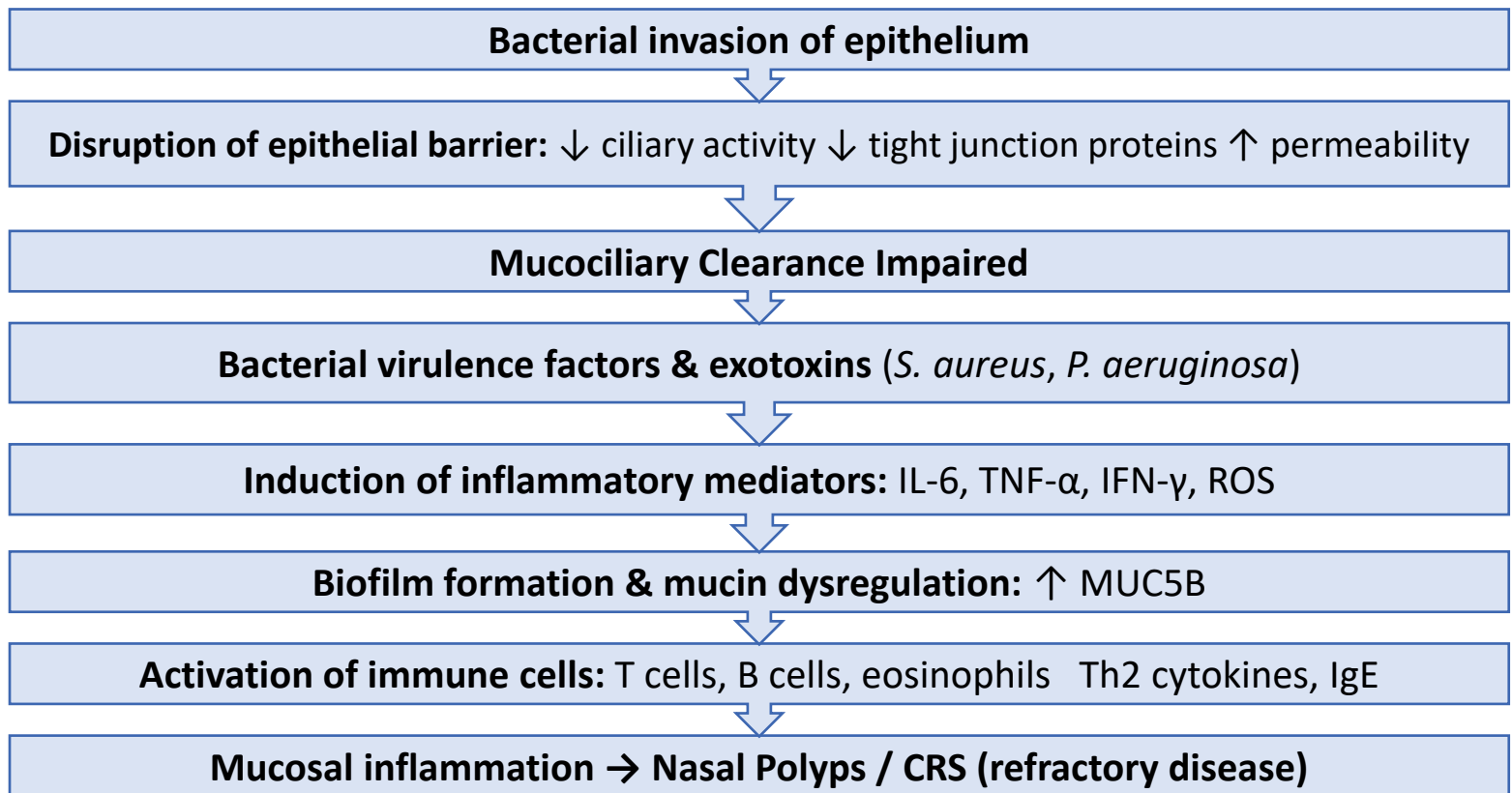
EPIDEMIOLOGY³

- ❑ Higher incidence in **women than men (1.9-fold)**
- ❑ Peak in **adults aged 45–74 years**
- ❑ More common in **adults than children** (sinuses not fully developed in children)

PATHOGENESIS OF BACTERIAL RHINOSINUSITIS

Molecular & Cellular Mechanisms - From Viral Trigger to Bacterial Complication

PATHOGENESIS PATHWAY⁴



Biofilm (found in 42–80% of CRS) contributes to antibiotic failure and disease persistence.⁴

DIAGNOSTIC CRITERIA FOR ABRS

ABRS is distinguished from viral ARS by any one of three clinical presentation criteria:³

- ❖ **CRITERION 1:** Persistent symptoms lasting >10 days without clinical improvement.
- ❖ **CRITERION 2:** Severe onset with fever >39°C and purulent nasal discharge or facial pain lasting ≥3–4 consecutive days.
- ❖ **CRITERION 3:** 'Double-sickening': initial improvement followed by new fever, headache, or increased nasal discharge ('double-sickening').

Note: Rule out allergic rhinitis if nasal obstruction and discharge coexist with itchy nose, sneezing, and watery eyes.³

CLINICAL CHALLENGE: ANTIBIOTIC MISUSE IN URTIs

Clinical, Economic, and Resistance Implications

Upper respiratory tract infections (URTIs), particularly acute rhinosinusitis, are predominantly viral and self-limiting; however, antibiotics continue to be widely prescribed without confirmed bacterial infection.³

In India, antibiotic use is common in primary care even without a definitive diagnosis of sinusitis, despite the self-limiting nature of the disease.³

~80%

of RTIs are caused by viruses — not bacteria⁵

≤75%

of antibiotics are prescribed for URTIs⁵

Key Drivers of Inappropriate Use

- **Diagnostic uncertainty:** Difficulty distinguishing viral vs bacterial infection³
- **Patient expectations:** Pressure influencing prescribing decisions^{3,5}
- **Practice variability across specialties³**
- **Misinterpretation of disease course:** Antibiotics often used despite symptoms <10 days³

CONSEQUENCES OF ANTIBIOTIC OVERUSE

Antimicrobial Resistance

- Promotes emergence of resistant pathogens and reduces antibiotic effectiveness^{3,5}

Clinical Impact

- Higher treatment failure rates and resistant infections⁵
- Antibiotic-related adverse events contribute to ~25% of ADRs in hospitalized patients⁵

Economic Burden

- Increased healthcare costs and resource utilization³
- Loss of productivity and impaired quality of life³

KEY TAKEAWAYS

Judicious antibiotic use is essential to reduce resistance, improve outcomes, and minimize unnecessary clinical and economic burden in URTIs

THERAPEUTIC RATIONALE — CO-AMOXICLAV

Addressing Antibiotic Resistance in Community-Acquired Respiratory Infections

- Amoxicillin-clavulanate is among the most frequently prescribed antimicrobial combinations in emergency departments and primary care offices worldwide.⁶
- Its position in community-acquired respiratory tract infection (RTI) management is defined by a single critical clinical problem: β -lactamase-mediated resistance renders amoxicillin monotherapy inadequate against key respiratory pathogens.⁶
- Clavulanate directly overcomes this resistance mechanism, restoring and broadening amoxicillin's clinical utility.⁶

The Clinical Problem → Rationale → Solution

The Problem

β -lactamase-producing organisms degrade amoxicillin's β -lactam ring, rendering it ineffective against key RTI pathogens⁶



The Rationale

Clavulanate binds and deactivates β -lactamases — preventing degradation of amoxicillin and keeping its antimicrobial mechanism intact⁶



The Solution

Co-amoxiclav delivers broad-spectrum coverage including β -lactamase-producing respiratory pathogens unreachable by amoxicillin alone⁶

Guideline-Endorsements

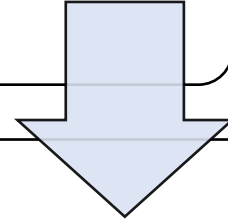
- ✓ **IDSA Guideline:** Preferred over amoxicillin alone for ABRS.⁶
- ✓ **AAO-HNS Guideline:** Recommends amoxicillin-clavulanate for ABRS.⁶
- ✓ **ATS-IDSA Guideline:** Recommended (as part of combination therapy) for CAP with comorbidities.⁶
- ✓ **ACG Guideline:** Suggested for SIBO (~50% efficacy).⁶
- ✓ **NHSRC India:** First-line empiric therapy for ABRS.³

MECHANISM OF ACTION

Amoxicillin–Clavulanate: Dual Mechanism of Antibacterial Activity

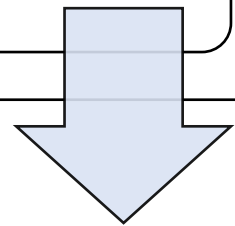
Amoxicillin — Bactericidal Action

- Amoxicillin, a broad-spectrum β -lactam penicillin, exerts its antibacterial effect by inhibiting bacterial cell wall synthesis through binding to penicillin-binding proteins (PBPs), thereby preventing peptidoglycan cross-linking.
- Inhibition of PBPs (DD-transpeptidases) disrupts cell wall integrity, ultimately leading to bacterial lysis and death.⁶



Bacterial Resistance — β -Lactamase

- Resistance to β -lactam antibiotics commonly arises through the production of β -lactamase enzymes, which hydrolyse the β -lactam ring and render amoxicillin ineffective.⁶



Clavulanic Acid — β -Lactamase Inhibitor

- Clavulanic acid is combined with amoxicillin to overcome this resistance mechanism; it binds to and irreversibly inactivates β -lactamases, protecting the β -lactam ring from degradation.
- This combination restores and extends the antimicrobial activity of amoxicillin against β -lactamase-producing organisms, thereby enhancing clinical effectiveness in community-acquired infections.⁶

EXTENDED COVERAGE WITH CLAVULANATE — KEY ORGANISMS

- All β -lactamase-producing strains of covered organisms⁶
- MSSA (methicillin-sensitive *S. aureus*)⁶
- *Neisseria* species⁶
- *Proteus* species⁶
- *Pasteurella multocida*⁶
- *Capnocytophaga canimorsus*⁶

FDA-APPROVED INDICATIONS (Amoxicillin-Clavulanate)

- Community-acquired pneumonia (CAP)⁶
- Acute bacterial rhinosinusitis (ABRS)⁶
- Acute otitis media (AOM)⁶
- Aspiration pneumonia⁶
- Urinary tract infection⁶
- Skin or soft tissue infection⁶

PK/PD BASIS OF AMOXICILLIN–CLAVULANATE XR

Time-Dependent Killing and the Rationale for Extended-Release

Clinical effectiveness of β -lactam antibiotics is determined by in vivo pharmacokinetic/pharmacodynamic (PK/PD) parameters, rather than in vitro susceptibility alone.⁷

Time-dependent killing — β -lactam mechanism

- Activity depends on the duration for which drug concentrations exceed the MIC ($T > \text{MIC}$), rather than AUC/MIC.⁷
- Optimal efficacy requires serum concentrations above MIC for ~40–50% of the dosing interval.⁷

Clinical Limitation of Conventional Regimens⁷

- Rising MICs among respiratory pathogens, particularly *Streptococcus pneumoniae*, reduce treatment effectiveness.
- Conventional dosing may fail to achieve adequate $T > \text{MIC}$, impacting bacterial eradication.

PK/PD Advantage of XR⁷

- Enhances $T > \text{MIC}$, the key determinant of β -lactam efficacy
- Maintains activity against:
 - β -lactamase-producing organisms
 - *S. pneumoniae* with reduced penicillin susceptibility (MIC up to 2 $\mu\text{g/mL}$)

Amoxicillin–clavulanate XR provides a PK/PD-driven strategy to overcome reduced susceptibility and optimize outcomes in community-acquired respiratory tract infections.⁷

FORMULATION DESIGN & CLINICAL RATIONALE

Amoxicillin/Clavulanate Extended-Release (ER) Bilayer Tablet

Rationale for XR Formulation⁷

- Amoxicillin–clavulanate XR is a bilayer, dual-release formulation.
- Provides immediate and sustained release components in a single bilayer tablet.
- Designed to optimize PK/PD exposure ($T > MIC$).

Bilayer Tablet Architecture⁷

Layer 1 - Immediate-release

- **Amoxicillin trihydrate 562.5 mg** — provides peak plasma concentration.
- **Clavulanate potassium 62.5 mg** — immediate release; β -lactamase inhibition from first dose.
- Enables early onset of antibacterial activity.

Layer 2 - Sustained-release

- **Crystalline amoxicillin sodium 437.5 mg** — sustained release.
- Maintains serum concentrations above MIC for a prolonged duration.
- Prolongs bacterial exposure to the antimicrobial.
- Supports improved $T > MIC$, the key PK/PD driver of β -lactam efficacy.

2000 mg

Amoxicillin per dose
(q12h)⁷

125 mg

Clavulanate per dose
(q12h)⁷

16:1

Amoxicillin:clavulanate
ratio⁷

Clinical Evidence⁷

In clinical studies of acute bacterial sinusitis (ABS) and community-acquired pneumonia (CAP):

- Demonstrated high clinical and bacteriological success rates.
- Generally well tolerated.

CLINICAL EVIDENCE — ACUTE BACTERIAL SINUSITIS (ABS)

TRIAL 1: Clinical & Radiological Effectiveness⁷

Design: Randomized, double-blind study (N = 363)

Regimen: Amoxicillin/clavulanate ER **2000 mg/125 mg BID for 10 days**

Endpoints

Clinical success: Improvement/resolution of symptoms without additional antibiotics

Radiological success: Unchanged or improved X-ray/CAT scan findings

Outcomes

Clinical success:

- **93.8%** at end of therapy
- **87.0%** at follow-up

Combined clinical + radiological success:

- **83.7%** at follow-up

TRIAL 2: Bacteriological Eradication & Clinical Success⁷

Design: Two open-label, noncomparative studies (Total N = 1554)

Bacteriological Outcomes

End of therapy: **96.9%** and **99.0%** eradication

Follow-up: **93.1%** and **96.4%** eradication

Clinical Outcomes

Clinical success: **>90%** at end of therapy and follow-up

Combined clinical + radiological success: **90–94%** of patients

Pathogen-specific eradication — *S. pneumoniae* with reduced penicillin susceptibility⁷

- *S. pneumoniae* overall eradication (ABS noncomparative studies): **97.7%**
- Strains with penicillin MIC = 2.0 µg/mL: **92.9% (13/14 patients)**
- Strains with penicillin MIC ≥ 4.0 µg/mL: **100% (9/9 patients)**
- *H. influenzae*: **94.1%** | *M. catarrhalis*: **98.4%**

Amoxicillin/clavulanate ER demonstrates consistently high clinical, radiological, and bacteriological success in ABS, including against pathogens with reduced penicillin susceptibility.⁷

CLINICAL EVIDENCE — COMMUNITY-ACQUIRED PNEUMONIA (CAP)

TRIAL 3 — Randomised, Double-Blind Study in CAP⁷

Parameter	Detail
Study Design	Randomised, double-blind clinical trial
Treatment Regimen	Amoxicillin/clavulanate ER 2000/125 mg twice daily × 7 or 10 days
Clinical Success at End of Therapy	90.5–96.0%
Clinical Success at Follow-Up	86.3–94.7%
Additional Outcomes	High bacteriological and radiological success documented

In this randomised, double-blind trial, amoxicillin/clavulanate ER achieved clinical success rates of 90.5–96.0% at end of therapy and 86.3–94.7% at follow-up in CAP.⁷

TRIAL 4 — Open-Label, Non-Comparative Study in CAP⁷

OVERALL OUTCOMES

Bacteriological Success (EOT)	87.0%
Bacteriological Success (Follow-Up)	83.0%
Clinical Success (EOT)	89.9%
Clinical Success (Follow-Up)	85.6%

S. PNEUMONIAE — SPECIFIC OUTCOMES

Outcome	Rate
Bact. success (EOT) ⁷	94.3%
Bact. success (Follow-Up) ⁷	91.6%
Clinical success (Follow-Up) ⁷	92.4%

Amoxicillin/clavulanate ER demonstrates consistently high clinical and bacteriological success in CAP, with robust efficacy observed across both randomized and real-world settings. High success rates at end of therapy and follow-up, including strong outcomes against *Streptococcus pneumoniae*, support its effectiveness in managing community-acquired pneumonia.⁷

TRIAL 5 — MULTI-INDICATION PK/PD EVIDENCE

Clinical Studies Supporting AMC/CA XR Across CAP, ABS, and AECS

Amoxicillin/clavulanic acid XR (2000/125 mg) maintains prolonged plasma levels of amoxicillin ($\geq 4 \mu\text{g/mL}$ for $\sim 49\%$ of dosing interval), improving activity against less susceptible *S. pneumoniae*.⁸

Indication	Study Type	Primary Outcome Measure	Clinical Success Rate
Community-Acquired Pneumonia (CAP) ⁸	Randomised Clinical Trials	Clinical success at test-of-cure	$\sim 86\text{--}95\%$
Acute Bacterial Sinusitis (ABS) ⁸	Clinical Studies	Combined clinical + radiological success (ITT)	$\sim 84\%$
Acute Exacerbation of Chronic Bronchitis (AECS) ⁸	Randomised Clinical Trials	Clinical success at test-of-cure	$\sim 83\text{--}93\%$

These outcomes support the clinical effectiveness of AMC/CA XR against *S. pneumoniae* with reduced susceptibility, driven by optimised PK/PD exposure.⁸

TRIAL 6 — SHORT-COURSE EFFICACY IN AECB

5-Day Pharmacokinetically Enhanced AMC/CA XR 2000/125 mg BD

Study Design & Population⁹

Parameter	Detail
Study Design	Randomised, controlled clinical study
Total Patients Randomised (N)	893
Patients Receiving AMC/CA XR	443
Treatment Regimen	AMC/CA XR 2000/125 mg twice daily × 5 days
Target Pathogens	Amoxicillin MICs ≤4 µg/mL, incl. β-lactamase producers
Test-of-Cure Assessment	Days 14–21 (per-protocol population)

The XR formulation was designed to maintain therapeutic amoxicillin serum concentrations over the 12-hour dosing interval.⁹

Key Efficacy & Tolerability Outcomes⁹

Outcome Measure	Result
Clinical Success (PP, test-of-cure) ⁹	93.0%
Bacteriological Success (bact-PP) ⁹	76.7%
Treatment Discontinuation (AEs) ⁹	<5%



TOLERABILITY

Favourable tolerability profile demonstrated: <5% of patients discontinued treatment due to adverse events in this 5-day short-course regimen.⁹

In this randomised, controlled study evaluating a 5-day course of AMC/CA XR 2000/125 mg twice daily in AECB, a clinical success rate of ~93% was observed in the per-protocol population at test-of-cure (days 14–21).⁹

Clinical Evidence Supporting Amoxicillin–Clavulanate XR

*Comparative Efficacy vs Conventional Comparators in Resistant *S. pneumoniae**

Study Design:

Pooled analysis of 9 clinical studies (6 randomized, double-blind; 3 non-comparative) in **CAP, ABS, and AECB**.¹⁰

Population:

ITT **N = 5531**, including patients with *S. pneumoniae* infections, including **penicillin-resistant strains (PRSP)**.¹⁰

Interventions:

AMX/CA XR 2000/125 mg BID

Comparators: Levofloxacin 500 mg OD, Clarithromycin 500 mg BID, AMX/CA 875/125 mg BID or 1000/125 mg TID.¹⁰

Key Clinical Advantages Summary¹⁰

- Overall success rates of **93.7–95.9%** were observed in *S. pneumoniae* respiratory infections.
- High efficacy (**95.2%**) demonstrated in isolates with **elevated MICs ≥ 4 $\mu\text{g/mL}$** .
- High success rate (**98.2%**) observed in **penicillin-resistant *S. pneumoniae* (PRSP)**.
- Comparator success rates were lower: **86.5% overall; 50% in PRSP cases**.
- Clinical effectiveness demonstrated across respiratory tract infections, including **ABS, AECB, and CAP**.
- Sustained **12-hour pharmacokinetic exposure** supports activity against resistant pathogens.

CLINICAL IMPLICATION:¹⁰

- AMX/CA 2000/125 mg BID is suitable for empirical treatment in regions with high antimicrobial resistance or in high-risk patients.
- Covers many PRSP strains non-susceptible to conventional amoxicillin $\pm$ clavulanate.

REGIONAL RELEVANCE — INDIA

Present Clinical Relevance of Amoxicillin–Clavulanate in CAP

- ❖ Antimicrobial resistance in CAP is a significant and growing concern in India, driven largely by inappropriate and excessive antibiotic use.¹¹
- ❖ Indian studies have reported notable resistance among respiratory pathogens, particularly to cotrimoxazole, erythromycin, and tetracycline.¹¹
- ❖ Despite this resistance landscape, amoxicillin–clavulanate retains high activity against major CAP pathogens including *S. pneumoniae*, *H. influenzae*, and *M. catarrhalis*.¹¹
- ❖ Indian and international susceptibility studies have reported high bacterial susceptibility, generally in the range of **>90% to 100%** for key respiratory pathogens.¹¹
- ❖ In Indian guidance, amoxicillin–clavulanic acid remains a first-line option for outpatient CAP, while in hospitalized CAP it is recommended as part of β -lactam plus macrolide combination therapy.¹¹
- ❖ The preference for combination therapy is based on broader pathogen coverage, inclusion of atypical pathogens, and potential mortality reduction.¹¹
- ❖ Ongoing dose evolution and PK/PD-optimized formulations reflect its continued relevance in current CAP practice.¹¹

FUTURE DIRECTIONS

Evolving Role of Amoxicillin–Clavulanate in CAP Management

- ❑ Amoxicillin–Clavulanate continues to play a key role in the management of community-acquired pneumonia (CAP), especially amid rising antimicrobial resistance and diagnostic uncertainty.¹²
- ❑ PK/PD optimization ($T > MIC \geq 40\text{--}50\%$) supports effective bacterial eradication.¹²
- ❑ The development of high-dose, bilayer, and pharmacokinetically enhanced formulations (including ES/XR) enables improved coverage against resistant pathogens while maintaining convenience through twice-daily dosing, thereby enhancing patient adherence.¹²
- ❑ Although amoxicillin–clavulanate does not provide coverage against atypical pathogens, it remains a key component of combination regimens with macrolides in CAP management.
- ❑ Demonstrates a favorable safety and tolerability profile, supporting widespread clinical use.¹²
- ❑ May have a potential role in selected resistant infections, warranting further exploration.¹²
- ❑ Antimicrobial stewardship and rational dosing will be critical to:
 - Maximize treatment outcomes
 - Limit resistance development
 - Ensure sustained long-term effectiveness

EXPERT CONSENSUS & PRESCRIBING TRENDS OF AMOXICILLIN–CLAVULANATE

Survey of 299 Physicians on URTI/LRTI Management¹³

Survey Parameter	Finding
URTI prevalence in practice	21–30% of patients
LRTI as most common antibiotic indication	83% of respondents
Most common URTI encountered	Pharyngitis (57%)
Bacterial aetiology estimated for URTIs	21–30% of cases
Basis for AMC 500/125 mg selection	Clinical history, comorbidities, likely pathogen, tolerability (63.55%)
AMC/CA preferred for routine RTIs	89% of surveyed physicians
AMC/CA preferred for paediatric AOM	88.29% of surveyed physicians
Preferred AMC/CA formulation for paediatric AOM	Syrup (86.29%)
Combination therapy preferred for acute sinusitis	83% of surveyed physicians

GUIDELINE-ENDORSED ROLE —ABRS & CAP

International and Indian Guideline Recommendations

IDSA Clinical Practice Guideline for ABRS:

- IDSA recommends amoxicillin–clavulanate over amoxicillin alone as initial empiric antimicrobial therapy for ABRS in both children and adults.¹⁴
- High-dose amoxicillin–clavulanate (ER) is recommended in patients at high risk of penicillin-non-susceptible *S. pneumoniae* infection.¹⁴

India — NHSRC Standard Treatment Guideline 2017:

- Amoxicillin-clavulanate is recommended as empiric antimicrobial therapy for ABRS in adults; recommended duration: 5–7 days.³
- High-dose amoxicillin-clavulanate (2 g twice daily) is recommended for patients with evidence of systemic toxicity, fever >39°C, age <2 or >65 years, recent hospitalisation, antibiotic use within the past month, or immunocompromised status.³

ATS–IDSA Community-Acquired Pneumonia Guidelines:

- ATS–IDSA CAP guidelines recommend amoxicillin–clavulanate–based regimens for outpatient adults with comorbidities.¹⁴
- Amoxicillin–clavulanate is preferred over respiratory fluoroquinolones as first-line empiric therapy in the outpatient comorbid population.¹⁴

Iran Guidelines:

- First-line empiric therapy for uncomplicated ABRS includes amoxicillin-clavulanate 500 mg/125 mg three times daily or 875 mg/125 mg twice daily.¹⁵
- In cases with high antibiotic resistance, high-dose amoxicillin-clavulanate 2000 mg/125 mg ER tablets twice daily is preferred.¹⁵

REGULATORY APPROVAL AND CLINICAL UTILITY OF AMOXICILLIN/CLAVULANATE XR

- ❑ Amoxicillin/clavulanate XR is a modified formulation designed to achieve and maintain therapeutic amoxicillin concentrations over an extended portion of the dosing interval, thereby optimising PK/PD exposure.⁷
- ❑ The formulation has demonstrated clinical efficacy in the treatment of acute bacterial sinusitis (ABS) and community-acquired pneumonia (CAP), with favorable clinical, bacteriological (confirmed or presumed), and radiological response rates reported in clinical studies.⁷
- ❑ Amoxicillin/clavulanate XR is active against *S. pneumoniae* with reduced penicillin susceptibility (MIC up to 2 µg/mL), as well as *H. influenzae* and *M. catarrhalis*, including β-lactamase–producing strains.⁷
- ❑ The XR formulation is particularly relevant in patients at risk of infection with penicillin-non-susceptible respiratory pathogens.¹⁴

Key takeaways



Guideline-endorsed first-line therapy Recommended by IDSA, ATS–IDSA, AAO-HNS, and NHSRC as first-line empiric therapy for ABRS and CAP with comorbidities; high-dose XR preferred in patients at risk of penicillin-non-susceptible *S. pneumoniae*.^{3,14}

PK/PD-optimized bilayer formulation Bilayer XR design maintains amoxicillin concentrations above MIC throughout the dosing interval, overcoming β -lactamase-mediated resistance including against *S. pneumoniae* with reduced penicillin susceptibility (MIC up to 2 $\mu\text{g/mL}$).⁷

High success rates in CAP and ABS Clinical success of 90.5–96.0% in CAP and 93.8% in ABS at end of therapy, with bacteriological eradication >93% against *S. pneumoniae*, *H. influenzae*, and *M. catarrhalis*.⁷

Superior efficacy against PRSP Achieved 98.2% success in penicillin-resistant *S. pneumoniae* vs only 50% with comparators, supporting use in high-resistance settings and high-risk patients.¹⁰

Well-tolerated with strong physician preference <5% treatment discontinuation due to adverse events; preferred by 89% of physicians for routine RTIs, supporting its role in rational antibiotic prescribing.^{9,13}

ABRS – Acute Bacterial Rhinosinusitis, ABS – Acute Bacterial Sinusitis, AECB – Acute Exacerbation of Chronic Bronchitis, ADRs – Adverse Drug Reactions, AAO-HNS – American Academy of Otolaryngology–Head and Neck Surgery, ACG – American College of Gastroenterology, AMC/CA – Amoxicillin/Clavulanic Acid, AMX/CA – Amoxicillin/Clavulanic Acid, AOM – Acute Otitis Media, ARS – Acute Rhinosinusitis, ATS – American Thoracic Society, AUC – Area Under the Curve, BID/BD – Twice Daily, CAP – Community-Acquired Pneumonia, CAT – Computed Axial Tomography, CRS – Chronic Rhinosinusitis, EOT – End of Therapy, ER – Extended-Release, ES – Extra Strength, FDA – Food and Drug Administration, ICU – Intensive Care Unit, IDSA – Infectious Diseases Society of America, ITT – Intent-to-Treat, LRTI – Lower Respiratory Tract Infection, MIC – Minimum Inhibitory Concentration, MSSA – Methicillin-Sensitive *Staphylococcus aureus*, NHSRC – National Health Systems Resource Centre, PK/PD – Pharmacokinetics/Pharmacodynamics, PP – Per Protocol, PRSP – Penicillin-Resistant *Streptococcus pneumoniae*, RTI – Respiratory Tract Infection, SIBO – Small Intestinal Bacterial Overgrowth, T>MIC – Time above Minimum Inhibitory Concentration, URTI – Upper Respiratory Tract Infection, WHO – World Health Organization, XR – Extended-Release

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